The second type of vowel harmony dictates the assimilation of roundness but also takes into account the closeness of the vowels. The possessive suffix for 1SG ('my') in Turkish has four allomorphs:[1] *-im*, *-ım*, *-um*, *-üm*. Similarly, the choice of suffix depends on the last vowel of the word, as follows:

- if the last vowel is front unrounded, use the form *-im*;
- if the last vowel is back unrounded, use the form *-ım*;
- if the last vowel is front rounded, use the form *-üm*;
- if the last vowel is back rounded, use the form *-um*.

Briefly, we can also represent this harmony as follows:

$$\{a, ı\} \longrightarrow ı \qquad \{e, i\} \longrightarrow i \qquad \{o, u\} \longrightarrow u \qquad \{ö, ü\} \longrightarrow ü$$

Therefore, if the last vowel is *a* or *ı*, use the vowel *ı*; if the last vowel is *o* or *u*, use the vowel *u*, and so on. We have the following pairs: *ev – ev**im***, *hortum – hortum**um***, *raf – raf**ım***, *göz – göz**üm***.

Moreover, notice that in Turkish all the vowels in a word have the same backness (the word only contains either front or back vowels). While this is a general trend, there are also exceptions to this rule, especially for words borrowed into Turkish from other languages.

Problem 4.4

Valley Yokuts (Paul Helmer, RoLO 2019)

Here are some verbal forms in Valley Yokuts in four different forms and their English translations:

Dubitative	Passive voice	Non-future	Imperative	Translation
do:sol	__(1)__	*doshin*	__(2)__	'to report'
__(3)__	*dubut*	*dubhun*	*dubk'a*	'to conduct'
yawa:lal	*yawa:lit*	*yawalhin*	*yawalk'a*	'to follow'
logwol	*logwit*	*logiwhin*	*logiwk'a*	'to pulverise'
wo:nol	*wo:nit*	*wonhin*	__(4)__	'to hide'

[1] In reality, there is also a fifth form, *-m*, for words ending in a vowel.

Dubitative	Passive voice	Non-future	Imperative	Translation
xatal	*xatit*	*xathin*	__(5)__	‘to eat’
__(6)__	__(7)__	__(8)__	*t’oyixk’a*	‘to treat’
__(9)__	*ʔopo:tit*	*ʔopothin*	*ʔopotk’o*	‘to get out of bed’
ʔugnal	__(10)__	*ʔugunhun*	__(11)__	‘to drink’
__(12)__	__(13)__	__(14)__	*ʔilikk’a*	‘to sing’
__(15)__	*lihmit*	*lihimhin*	__(16)__	‘to run’
__(17)__	*luk’lut*	__(18)__	__(19)__	‘to bury’
__(20)__	*k’oʔit*	__(21)__	*k’oʔk’o*	‘to throw’
me:k’al	__(22)__	__(23)__	__(24)__	‘to swallow’

Problem 4.4a Fill in the blanks. If you believe that some blanks could allow multiple answers, write them all.



k’, *t’*, *x*, *y*, *ʔ* are consonants. The mark *:* after a vowel denotes length.

Solution

We begin by segmenting the forms in order to figure out which part is the stem and which are the morphemes corresponding to the four verb forms. We notice that, for two of the verbs, we are given all four forms:

Dubitative	Passive voice	Non-future	Imperative	Translation
yawa:lal	*yawa:lit*	*yawalhin*	*yawalk’a*	‘to follow’
logwol	*logwit*	*logiwhin*	*logiwk’a*	‘to pulverise’

Comparing these forms, we deduce that the dubitative is formed by adding the suffixes *-al* or *-ol*, the passive voice is formed using the suffix *-it*, the non-future using *-hin* and the imperative using *-k’a*. Moreover, we notice that the stems can undergo some changes (for the verb ‘to follow’ there are two possible stems *yawa:l* and *yawal*, while for the verb ‘to pulverise’ there is *logw* and *logiw*). Moreover, we notice that, in both cases, the dubitative and passive voice use the same stem, while the other two forms use the “modified” stem.